

Rings and Modules

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January 22, 2026

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Chapter 1

Rings: Definition and Examples

Definition 1. A ring is a datum $(R, +, \times, 0, 1)$ where R is a set, $1, 0 \in R$ and $+, \times$ are binary operations on R such that

- R is an abelian group under $+$ with identity element 0
- The binary operation $\times$ is associative and $1 \times x = x \times 1 = x$ for all $x \in R$.
- Multiplication distributes over addition:

$$\begin{aligned}x \times (y + z) &= (x \times y) + (x \times z), \\(x + y) \times z &= (x \times z) + (y \times z), \forall x, y, z \in R.\end{aligned}$$

Example. Common rings:

- The integers $\mathbb{Z}$ are the fundamental example of a ring. Much of the course will be about finding an interesting class of rings which behave a lot like $\mathbb{Z}$.
- $\mathbb{Z}[i] = \{a + ib \in \mathbb{C} : a, b \in \mathbb{Z}\}$ is a ring. It is known as the *Gaussian integers*.
- Any field is a ring. The only difference between the axioms for a field and for a ring is that a ring does not require multiplicative inverses, and that for fields one assumes $1 \neq 0$. Also fields must have commutative multiplication, but we assume it for rings in this course.

Lemma 1. (Subring criterion) Let R be a ring and S a subset of R , then S is a subring if and only if $1 \in S$ and for all $s_1, s_2 \in S$ we have $s_1 s_2, s_1 - s_2 \in S$.

Chapter 2

Basic Properties

2.1 Integral Domains

Definition 2. Let R be a ring. The subset

$$R^\times = \{r \in R : \exists s \in R, rs = 1\}$$

is called the group of *units* in R , i.e. $(R, \times, 1_R)$ is a group. Intuitively these are the invertible elements.

Proposition 1. Let K be a field and let $\theta : R \rightarrow K$ be an embedding (injective homomorphism). Then there is a unique injective homomorphism $\tilde{\theta} : F(R) \rightarrow K$ extending θ , in the sense that $\tilde{\theta}|_R = \theta$ where we view R as a subring of $F(R)$ via the above embedding.

Remark. If $u \in I$ where $u \in R^\times$, then $uu^{-1} \in I$, then $1_R \in I$, then $I = R$.